

06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel
cmp-legacy.`

The business decision. Brand search converts like crazy — should we pour budget into it? Be careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created, so crediting it for those conversions and shifting budget its way would be a mistake.

1.0.1 The concepts

- **MMM (Marketing Mix Model)** — a regression of sales on each channel’s spend over time (usually weekly), used to attribute sales to channels and set budgets. It’s the “top-down” alternative to click-tracking.
- **Adstock (carry-over)** — advertising doesn’t only work the week it runs; its effect *decays over subsequent weeks*. Adstock transforms raw spend into an “effective” exposure that carries over.
- **Saturation (diminishing returns)** — the 10th €1,000 into a channel buys less lift than the 1st. A saturation curve bends over as spend rises; its *slope* is the **marginal** return that should drive the next euro of budget.
- **Last-click attribution** — crediting the conversion to the final touch (often brand search). It systematically **over-credits** demand-capture channels, which is the error a causal MMM aims to fix.
- **Incremental vs mediated** — the *incremental* contribution of a channel is what it *caused*; a channel can show a big correlation with sales while causing very little (it was downstream of the real driver).

1.0.2 The arc of this notebook — a cautionary tale with a resolution

A plain MMM regresses sales on all channels and is fooled by a **confounder** like seasonality (it lifts sales *and* drives brand-search spend). The textbook fix is a **causal MMM**: hand it a **DAG** and let the backdoor logic (nb 05) control for the confounder. This notebook shows that fix is **necessary but not sufficient**, in three acts:

1. **The confounding correction works — in a linear model.** A naive OLS attribution that omits seasonality massively over-credits `brand_search`; adding seasonality as a control collapses that credit toward the small truth. That’s the cleanest way to *see* the confounding, and it’s robust.

2. **The out-of-the-box causal MMM still inverts the ranking.** Fit with library-default priors, the MMM’s *flexible saturation curve* for `brand_search` **re-absorbs** the seasonal demand the linear control was meant to remove — so it credits `brand_search` *more* than TV, the opposite of the truth. A DAG plus a linear control does **not** de-confound a channel whose spend is collinear with the confounder.
3. **Experiment-calibrated priors recover it.** Feed in what a small **geo experiment** teaches — that `brand_search`’s incremental effect is small — as an informative prior on its saturation, and the MMM ranks the channels correctly again. **This is why MMMs must be calibrated with experiments** (Anchor B, nb 07): observational data alone cannot pin the decomposition.

On real data. Swap in your **own weekly spend-by-channel + sales** table plus known confounders (seasonality, price, distribution). Public example: the datasets shipped with Meta’s *Robyn* or `pymc-marketing`’s own tutorials. Always pair an MMM with occasional geo experiments to anchor the levels.

It follows the repo’s fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Weekly sales from a faithful **adstock + Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave (corr ~ 0.75), so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, `brand_search` only slightly.

The data-generating model — exactly what `dgp.mmm_weekly` implements (defaults & seed in `src/cmp/dgp.py`). Weeks $t = 0, \dots, 155$; seasonality $s_t = \sin(2\pi t/52)$:

$$\begin{aligned}
 x_t^{\text{tv}} &= \max\{1, 40 + 8 s_t + \varepsilon_t^{\text{tv}}\}, & \varepsilon_t^{\text{tv}} &\sim \mathcal{N}(0, 8^2); & x_t^{\text{bs}} &= \max\{1, 25 + 11 s_t + \varepsilon_t^{\text{bs}}\}, & \varepsilon_t^{\text{bs}} &\sim \mathcal{N}(0, 7^2) \\
 a_t(x; \lambda) &= x_t + \lambda a_{t-1}(x; \lambda) & & \text{(geometric adstock / carry-over)} \\
 f(a; \alpha, \kappa) &= \frac{\alpha a}{\kappa + a} & & \text{(Michaelis-Menten saturation)} \\
 \text{sales}_t &= 100 + \underbrace{f(a_t(x^{\text{tv}}; 0.4); 60, 50)}_{\text{TV: big, real}} + \underbrace{f(a_t(x^{\text{bs}}; 0.2); 8, 40)}_{\text{brand search: small, real}} + \underbrace{35 s_t}_{\text{season} \rightarrow \text{sales}} + \varepsilon_t, & & \varepsilon_t &\sim \mathcal{N}(0, 4^2).
 \end{aligned}$$

The confounding is the **two appearances of s_t** : it drives brand-search spend (the $11 s_t$ term — people search the brand more in peak season; corr ~ 0.75) *and* sales directly (the $35 s_t$ term). An MMM that omits seasonality hands much of that $35 s_t$ to brand search. Note the generator uses the MMM’s own functional form (adstock -> saturation), so the model is well-specified — any remaining failure is about *identification*, not fit.

TRUE incremental sales - TV 5,263 brand_search 518 (small)

	date_week	tv	brand_search	seasonality	sales
0	2023-01-02	42.870187	35.579322	0.000000	134.900978
1	2023-01-09	53.049712	17.732754	0.120537	137.863335

2	2023-01-16	27.623875	23.486826	0.239316	142.263680
3	2023-01-23	56.329747	34.885672	0.354605	159.788875
4	2023-01-30	43.339248	33.736743	0.464723	152.485792

1.2 3 · Identify — the marketing DAG picks controls vs mediators

DAG: seasonality → {sales, brand_search}, tv → sales, brand_search → sales. The backdoor criterion says **seasonality is a confounder → control for it**; omitting it is exactly the last-click mistake that over-credits brand search. The estimand for channel c is its **incremental contribution** — sum over weeks of $\mathbb{E}[\text{sales} \mid \text{do}(\text{spend}_c = \text{observed})] - \mathbb{E}[\text{sales} \mid \text{do}(\text{spend}_c = 0)]$, holding the other channels at their observed spend (how much of sales would vanish if we switched channel c off). ROI = contribution ÷ spend, each with a posterior. Identification comes from the backdoor set {seasonality}; we hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.

Backdoor adjustment set the causal layer will use: ['seasonality']

1.3 4 · Estimate — fit the causal MMM twice

We fit the **same DAG-aware MMM twice**, to make the lesson unmissable:

- (a) **default priors** — the honest, out-of-the-box causal MMM. Watch it *invert* the ranking.
- (b) **experiment-calibrated priors** — encode what a geo experiment teaches (brand_search’s incremental effect is small) as an informative prior on its saturation ceiling, and watch the ranking snap back.

The fitted model, in symbols. Per channel $c \in \{\text{tv}, \text{brand_search}\}$, spend is adstocked over an 8-week window, saturated, and summed with the control:

$$\text{sales}_t \sim \mathcal{N}\left(\beta_0 + \sum_c f_c(a_c(x^c)_t) + \gamma s_t, \sigma^2\right), \quad a_c(x)_t \propto \sum_{\ell=0}^7 \lambda_c^\ell x_{t-\ell}, \quad f_c(a) = \frac{\alpha_c a}{\kappa_c + a},$$

geometric adstock (`GeometricAdstock(l_max=8)`) and Michaelis–Menten saturation, with each channel’s decay λ_c , ceiling α_c and half-saturation κ_c **estimated** from the data. Fit (a) leaves their priors at library defaults; fit (b) replaces the prior on brand-search’s ceiling α_{bs} with the experiment-informed one. Note the functional form matches step 2’s generator exactly — this is the *well-specified best case*, which is what makes the ranking inversion in (a) a lesson about identification rather than model misfit. (pymc-marketing scales variables internally; the equations are written in original units for readability.)

Convergence housekeeping: this is the hardest posterior geometry in the cookbook — the FULL profile needs `target_accept = 0.99` and deeper trees to reach R-hat ~ 1.01 (the diagnostics — R-hat, ESS, divergences — are defined in notebook 01’s “did the sampler converge?” box).

A caution that runs through both: **MMMs are miscalibrated in absolute terms** — with channel spend riding seasonal demand, a short weekly series can’t fully separate “channel drove sales” from “season drove both”, so fitted contributions run high. What we’re after is the **ranking** and each channel’s **ROI verdict**.

Output()

```
(a) default-prior MMM convergence: max r-hat 1.010 - min ESS 381 - divergences
14
(a) default contribution - TV 10,390    brand_search 36,007
    true: TV 5,263, brand_search 518 -> the MMM ranks brand_search ABOVE TV
(inverted!).
```

1.3.1 Why the out-of-the-box MMM inverts the ranking

The default MMM credits `brand_search` *above* TV — backwards. The mechanism is worth seeing, because it’s a real trap:

- `brand_search`’s spend is *built* to ride the seasonal demand wave (Step 2 constructs it from the same seasonal index that drives sales), and the fit lets its flexible saturation curve quietly **re-absorb the seasonal demand** the linear **seasonality** control was meant to remove.
- Because `brand_search` rides the seasonal wave more tightly than TV does, it wins that tug-of-war and books a huge contribution (tens of thousands vs a true ~500), while a large **negative baseline** offsets it — the classic non-identified MMM level decomposition.
- The DAG was right and the control was present; they still weren’t enough. **A linear confounder-control does not de-confound a channel whose spend is collinear with the confounder and enters through a flexible functional form.** Observational data alone can’t break the tie.

1.3.2 The fix — priors calibrated from a geo experiment

The standard remedy is exactly the discipline of Anchor B (nb 07): run a **small geo experiment**, learn that `brand_search`’s *incremental* effect is small, and feed that back as an **informative prior on its saturation ceiling** (`saturation_alpha`). TV, whose incremental role we haven’t independently pinned down, keeps a weakly-informative prior. This is not putting a thumb on the scale — it’s using experimental evidence exactly where observational data is silent.

Output()

```
(b) calibrated-prior MMM convergence: max r-hat 1.010 - min ESS 376 -
divergences 0
(b) calibrated contribution - TV 12,820    brand_search 2,262
    true: TV 5,263, brand_search 518 -> ranking RESTORED: TV above
brand_search.
```

1.4 5 · Validate — the correction, the recovered ROI, and saturation

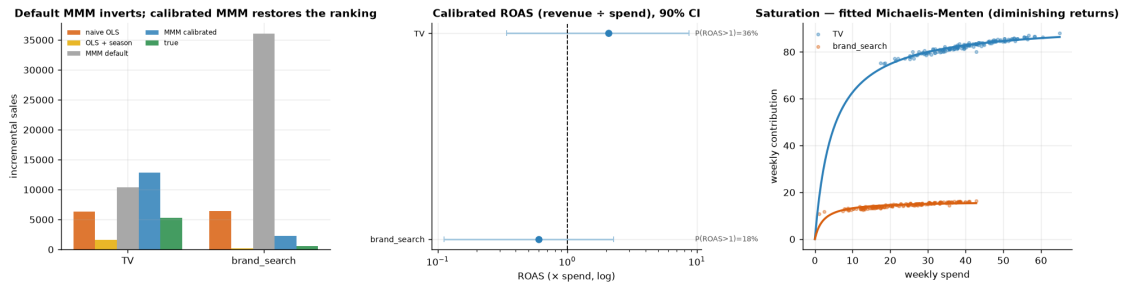
Three views:

1. **The confounding correction, several ways** — naive OLS (omit season) over-credits `brand_search`; OLS + **season** collapses it toward truth; the **default** causal MMM *inverts* it; the **calibrated** MMM restores the right ranking. The contrast between the two MMMs is the headline.
2. **ROI with uncertainty (calibrated MMM)** — each channel’s posterior ROI vs the break-even line. $ROAS = \text{contribution} \div (\text{fixed observed spend})$, so it carries the **same relative**

miscalibration as the contribution — dividing by a constant spend cannot remove the bias. What is robust is the **ROAS ranking** (and `brand_search`’s below-break-even verdict), **not** the absolute ROAS *levels*, which are biased by the same factor as the contributions.

3. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that sets *marginal* ROI and hence where the next euro should go).

```
brand_search credit - naive 6,401 | OLS+season 222 | MMM default 36,007 | MMM
calibrated 2,262 | true 518
calibrated ROAS - TV 2.08x [90% 0.34,8.67] · brand_search 0.60x [90%
0.11,2.28]
TRUE ROAS          - TV 0.85x · brand_search 0.14x (both below 1x: the MMM
recovers the ranking but overstates levels)
```



How to read this. *Left* tells the whole story in one panel. **Naive OLS** (orange) over-credits `brand_search`; **OLS + season** (gold) collapses that credit toward the small truth — the confounding correction, working. Then the two MMMs: the **default** causal MMM (grey, the failure mode) *inverts* the ranking, crediting `brand_search` far above TV — the seasonal-demand-stealing trap — while the **calibrated** MMM (blue) restores TV above `brand_search`, landing beside the **true** bars (green). *Middle* — the decision-relevant **calibrated ROI**: TV’s mean clears break-even but its 90% interval is wide and dips below 1x, while `brand_search`’s mean sits **below break-even**. So TV is the better channel, but its above-break-even verdict is itself uncertain. And the ranking is only *probably* right: draw-by-draw, TV’s ROI beats `brand_search`’s about **78%** of the time in this run (the $P(\text{TV ROI} > \text{brand_search ROI})$ computed next) — more likely than not, but **short of the 0.8 bar** we’d want before moving real budget. *Right* — the **saturation curves**: contribution flattens as spend rises (diminishing returns), so the *next* euro into a channel is worth its curve’s *slope*, not its average. The honest caveat (Step 7): even the calibrated ROI *levels* are approximate — anchor them with the same kind of geo experiment that supplied the prior.

1.5 6 · Decide, in euros — should we reallocate budget yet?

Budget chases **incremental return**, not last-click credit. One naming point first: what the MMM gives us is **ROAS** — incremental *revenue* per euro of spend — not profit ROI. Break-even at 1x therefore assumes a 100% margin; at a realistic margin m the break-even ROAS is $1/m$ (a 30% margin $\rightarrow$ 3.3x), which we show below.

Now that experiment-calibrated priors have made the *ranking* trustworthy, the point estimates put TV above `brand_search`, so the *direction* of any reallocation is toward TV. But a direction is not a decision, and the **absolute levels are still miscalibrated**. So we quantify three things: (1) the

sales impact of shifting a slice of budget — swept across shift sizes and made **saturation-aware**, so marginal ROAS falls as we pour more into TV; (2) the posterior probability the move actually helps; and (3) the honest gap between the MMM's ROAS and the *true* ROAS. That last check is the sobering one: it is why, even with the ranking fixed, the absolute “is TV worth it?” question still belongs to a geo test (Anchor B, nb 07), not a still-approximate MMM.

TV ROAS - average 2.08x · marginal (the next euro, at today's spend) 0.23x - marginal is lower because of saturation (diminishing returns).

Shifting 15% of brand_search spend (€566) to TV -> +€61 net sales (saturation-aware); the sweep peaks near 22%.

$P(\text{TV ROAS} > \text{brand_search ROAS}) = 0.78$ -> lean toward TV, but test first

Ranking vs levels - the watertight part:

TRUE ROAS: TV 0.85x · brand_search 0.14x (both BELOW 1x) | MMM says TV 2.08x · bs 0.60x

The MMM recovers the RANKING (TV >> brand_search) but overstates LEVELS ~2.4x for TV.

ROAS is revenue ÷ spend - it assumes 100% margin. At a realistic 30% margin the break-even is $1/0.30 \sim 3.3x$,

which even TV's estimated 2.08x fails; and TV's TRUE ROAS 0.85x is below break-even at ANY margin.

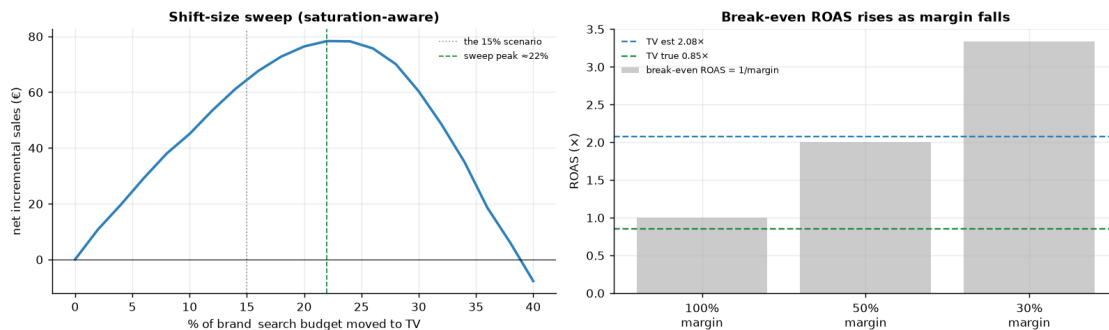
So: reallocate toward TV (the ranking is trustworthy), but 'is TV worth it in absolute €?' needs the geo

experiment (nb 07) - this MMM's absolute levels are not calibrated enough to answer it.

Decision quantity WITH uncertainty - shift 15% (€566) brand_search -> TV (per-draw, average-ROAS):

Deltasales = €836 [90% €-799, €4,618] · $P(\Delta > 0) = 0.78$

TRUE Deltasales (from DGP true_contrib) = €405 -> direction check: MATCH (both favour TV)



Which Deltasales number goes to the CFO? Two figures print above for the same 15% shift, and they differ ~13x on purpose: **+€61/week** is the *saturation-aware* estimate — it walks both channels down their fitted response curves, and is the honest planning number; **+€836/week**

prices the shift at *average* ROAS per posterior draw — it ignores saturation entirely and is kept only because it carries the uncertainty band (its $P(\Delta > 0) \sim 0.78$ is the ranking probability in euro clothing — by construction $\Delta > 0$ exactly when TV's ROAS beats brand_search's). Quote €61; use €836's band as the direction check it is.

1.6 7 · Caveats

- **Absolute contributions need calibration.** Even the calibrated MMM runs a fewx high in absolute euros, because channel spend rides the seasonal wave the model can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — and note it took an experiment-informed prior to get even the *ranking* right. This is the single most important thing to say about any MMM.
- **A DAG is necessary but not sufficient.** The correct backdoor control (seasonality) did not stop a flexibly-saturated, collinear channel from re-absorbing the confounder. Structural assumptions still need experimental calibration to be trustworthy.
- **Adstock/saturation priors matter — a lot.** Carry-over (λ) and diminishing-returns (α , κ) are weakly identified from limited weekly data; here the saturation-ceiling prior *flipped the conclusion*. Use informative priors from experiments wherever you have them.
- **Correlated channels -> wide, correlated ROI posteriors.** When spend moves with a confounder, the data can't fully separate them; the intervals will (honestly) show it.